F FWallSim · Firewall Course

CLASS 2

Your first firewall policy: LAN to Internet

Firewall policy, NAT, implicit deny

Beginner | ~15 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

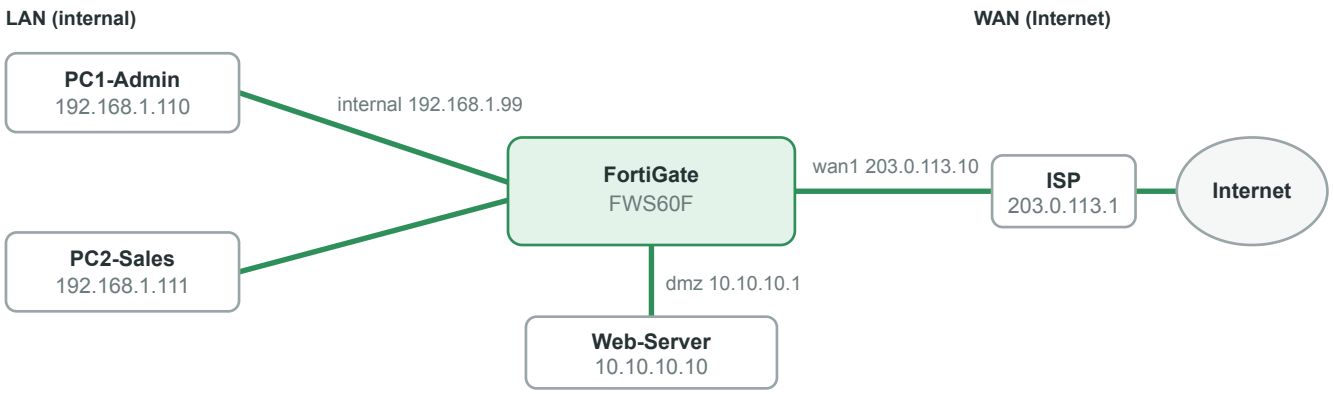
YOU WILL LEARN

- Firewall policy
- NAT
- implicit deny

BEFORE YOU START

Finish Class 1 first.

The lab network



The goal

THE IDEA

A new FortiGate blocks everything through it (the Implicit Deny). Build the policy that lets the office reach the internet, with source NAT so replies can come back.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **Your first firewall policy: LAN to Internet** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **Create an ACCEPT policy from internal to wan1**

   Policy & Objects > Firewall Policy > Create New. Incoming internal, Outgoing wan1, Source all, Destination all, Schedule always, Service ALL, Action ACCEPT.

2. **Enable NAT on that policy**

   In the same policy, Firewall / Network Options > NAT: on (Use Outgoing Interface Address).

3. **PC1 can ping 8.8.8.8**

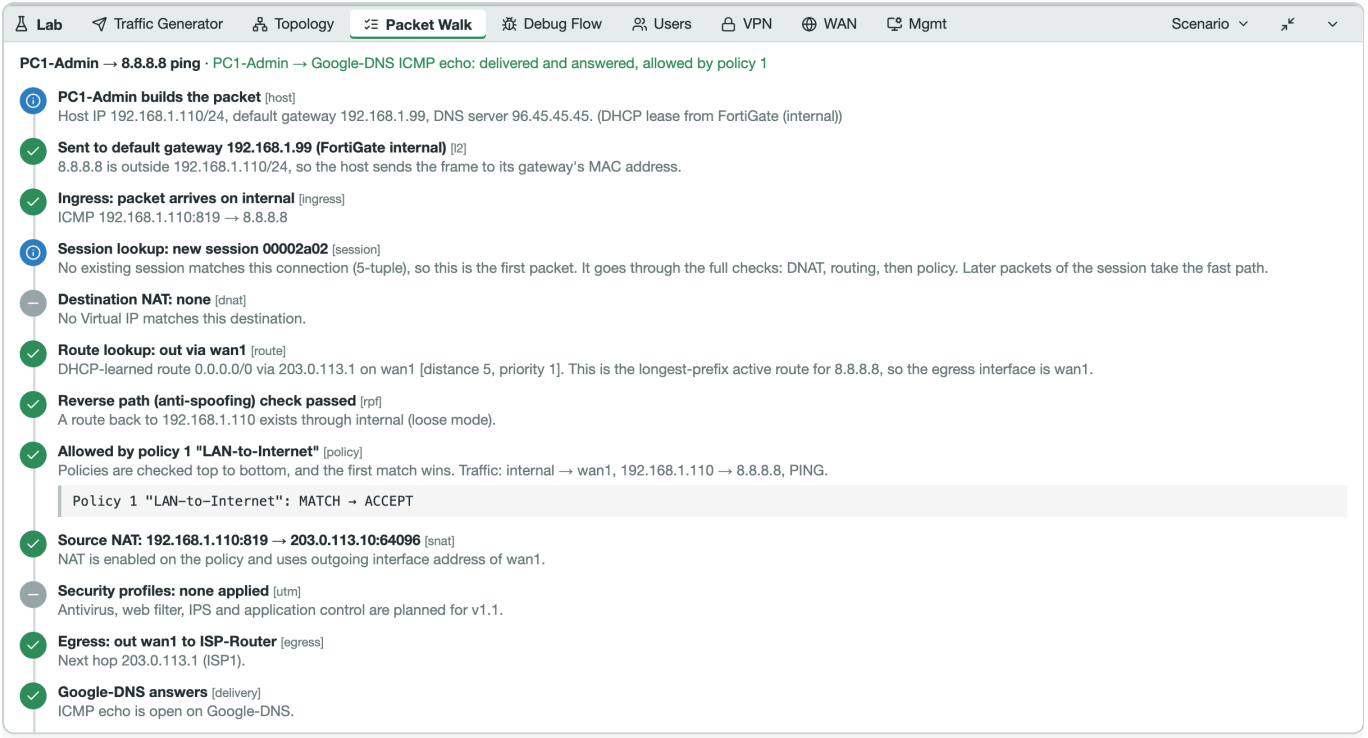
   Lab > Traffic Generator > quick test "PC1 pings 8.8.8.8". Look at the Source NAT step in the Packet Walk.

4. **PC1 can browse www.example.com**

   Quick test "PC1 browses www.example.com". This needs DNS first. The Packet Walk shows the DNS query too.

Screenshots

Numbered red markers show where to click or what to look at.



Allowed Walk

Lab | Traffic Generator | Topology | Packet Walk | **Debug Flow** | Users | VPN | WAN | Mgmt | Scenario

```
FWS60F # diagnose debug flow filter clear
FWS60F # diagnose debug flow show function-name enable
FWS60F # diagnose debug flow trace start 10
FWS60F # diagnose debug enable

id=65308 trace_id=2 func=print_pkt_detail line=5895 msg="vd-root:0 received a packet(proto=1, 192.168.1.110:819->8.8.8.8:2048) tun_id=0.0.0.0 from internal. type=8,
code=0, id=819, seq=1."
id=65308 trace_id=2 func=init_ip_session_common line=6076 msg="allocate a new session-00002a02, tun_id=0.0.0.0"
id=65308 trace_id=2 func=vf_ip_route_input_common line=2605 msg="find a route: flag=04000000 gw-203.0.113.1 via wan1"
id=65308 trace_id=2 func=fw_forward_handler line=881 msg="Allowed by Policy-1: SNAT"
id=65308 trace_id=2 func=__ip_session_run_tuple line=3497 msg="SNAT 192.168.1.110->203.0.113.10:64096"
id=65308 trace_id=3 func=print_pkt_detail line=5895 msg="vd-root:0 received a packet(proto=1, 8.8.8.8:0->203.0.113.10:2048) tun_id=0.0.0.0 from wan1. type=0, code=
0, id=0, seq=1."
id=65308 trace_id=3 func=resolve_ip_tuple_fast line=5976 msg="Find an existing session, id-00002a02, reply direction"
id=65308 trace_id=3 func=__ip_session_run_tuple line=3511 msg="DNAT 203.0.113.10:64096->192.168.1.110:819"
id=65308 trace_id=3 func=vf_ip_route_input_common line=2605 msg="find a route: flag=00000000 gw-192.168.1.110 via internal"

FWS60F # diagnose debug disable
```

Debug Flow

Lab | Traffic Generator | Topology | **Packet Walk** | Debug Flow | Users | VPN | WAN | Mgmt | Scenario

PC1-Admin → 8.8.8.8 ping · Denied by implicit deny: no policy allows internal → wan1 PING

PC1-Admin builds the packet [host]
Host IP 192.168.1.110/24, default gateway 192.168.1.99, DNS server 96.45.45.45. (DHCP lease from FortiGate (internal))

Sent to default gateway 192.168.1.99 (FortiGate internal) [l2]
8.8.8.8 is outside 192.168.1.110/24, so the host sends the frame to its gateway's MAC address.

Ingress: packet arrives on internal [ingress]
ICMP 192.168.1.110:117 → 8.8.8.8

Session lookup: new session 00002a01 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.

Destination NAT: none [dnat]
No Virtual IP matches this destination.

Route lookup: out via wan1 [route]
DHCP-learned route 0.0.0.0/0 via 203.0.113.1 on wan1 [distance 5, priority 1]. This is the longest-prefix active route for 8.8.8.8, so the egress interface is wan1.

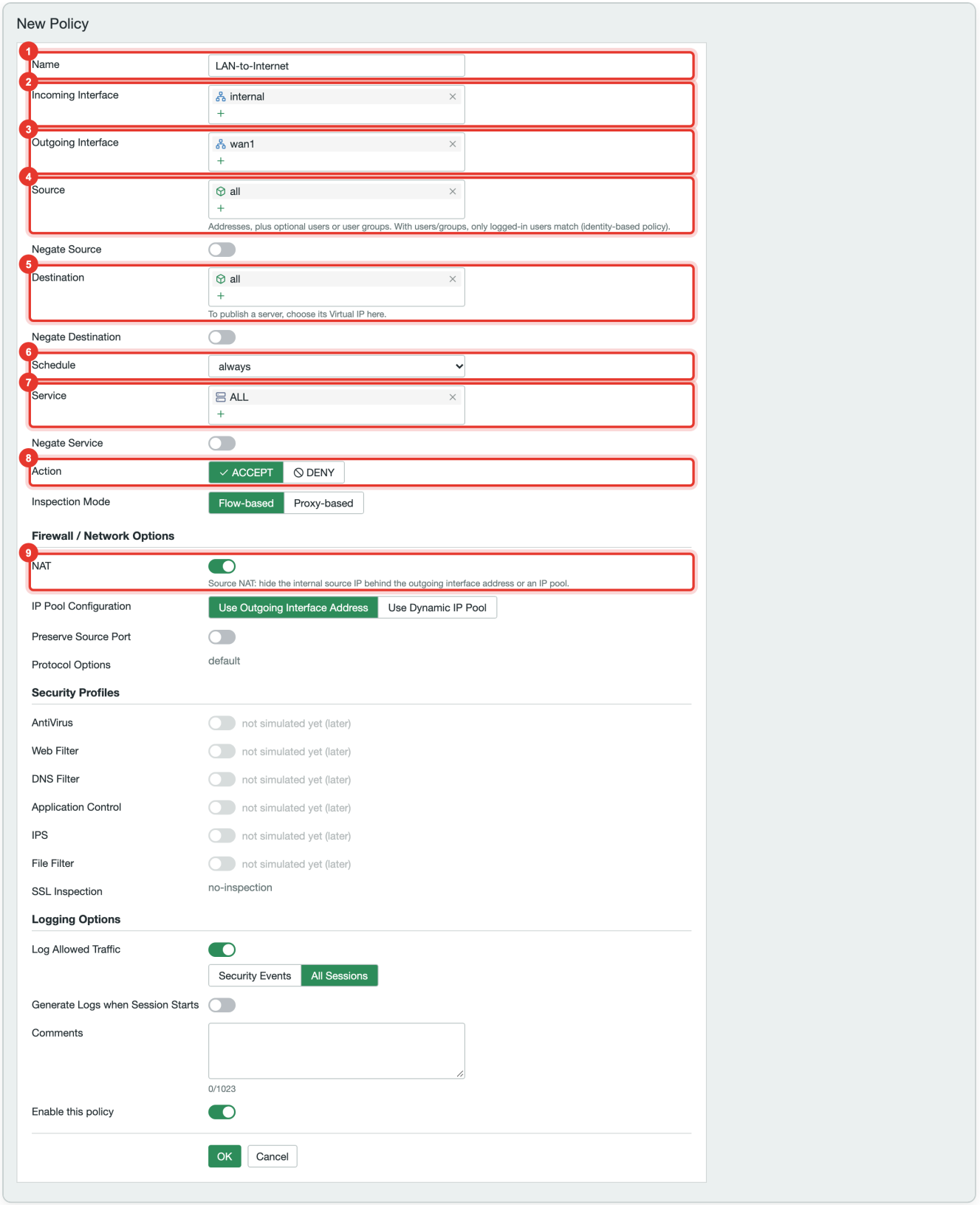
Reverse path (anti-spoofing) check passed [rpf]
A route back to 192.168.1.110 exists through internal (loose mode).

Denied by Implicit Deny (policy 0) [policy]
Policies are checked top to bottom, and the first match wins. Traffic: internal → wan1, 192.168.1.110 → 8.8.8.8, PING.

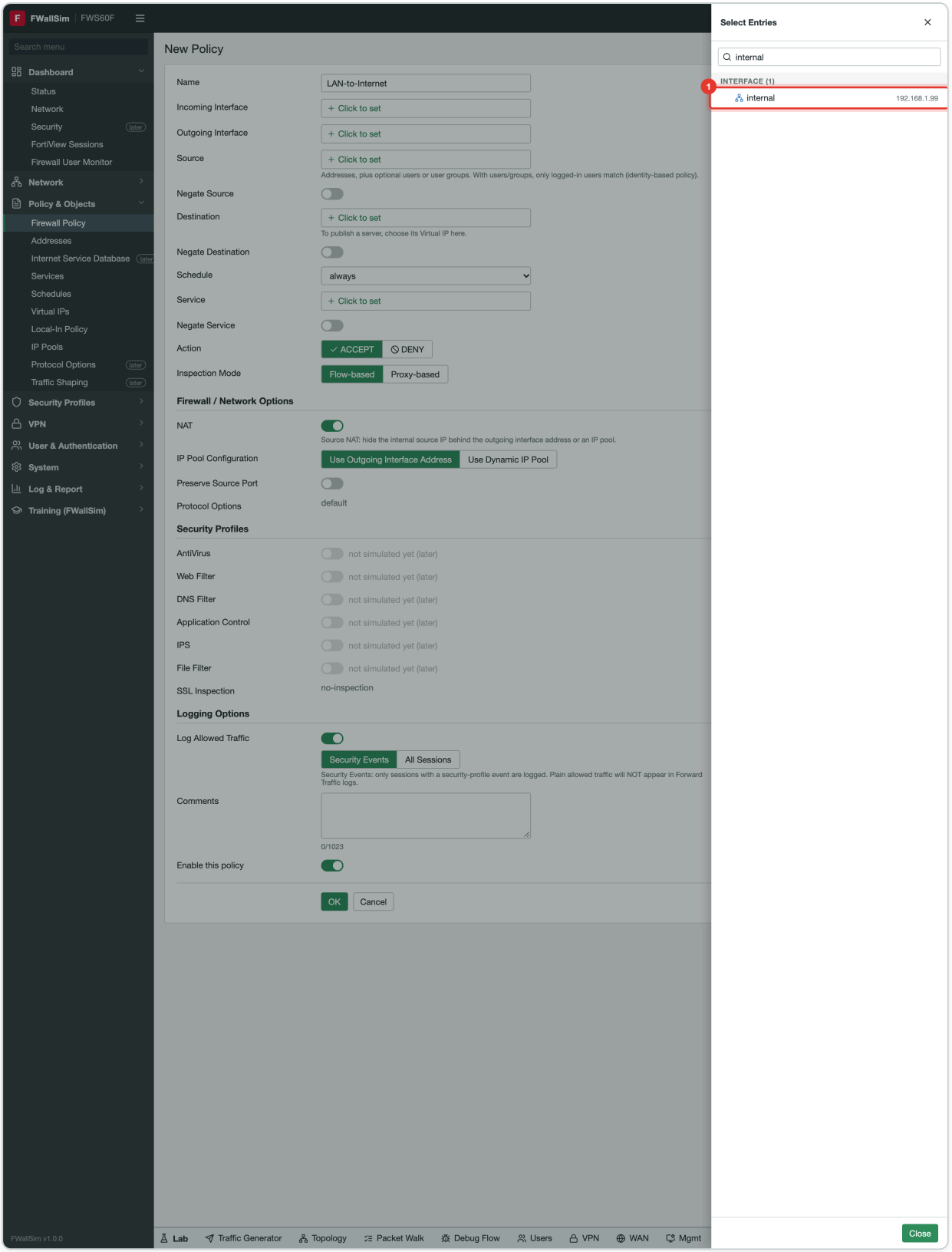
```
Nothing matched, so the Implicit Deny (policy 0) applies.
```

💡 Create a policy from internal to wan1 that matches this traffic (Policy & Objects > Firewall Policy).

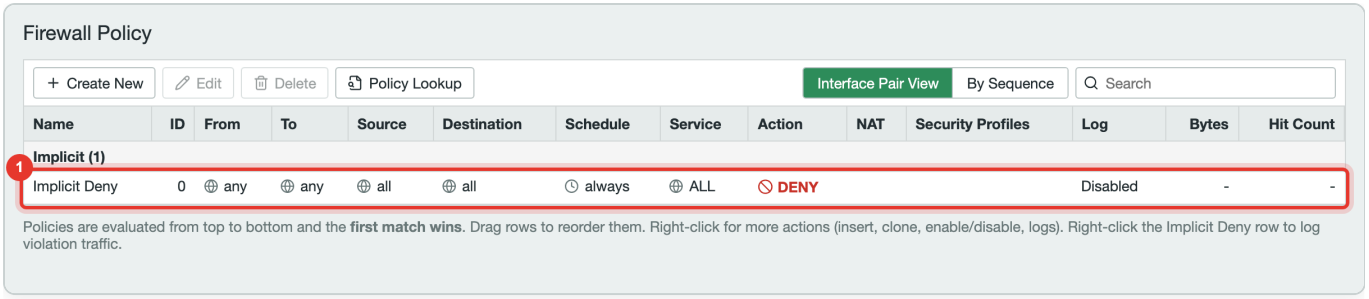
Denied Walk


New Policy
1
Name
LAN-to-Internet
2
Incoming Interface
internal
3
Outgoing Interface
wan1
4
Source
all
Addresses, plus optional users or user groups. With users/groups, only logged-in users match (identity-based policy).
Negate Source
5
Destination
all
To publish a server, choose its Virtual IP here.
Negate Destination
6
Schedule
always
7
Service
ALL
Negate Service
8
Action
ACCEPT
DENY
Inspection Mode
Flow-based
Proxy-based
Firewall / Network Options
9
NAT
Source NAT: hide the internal source IP behind the outgoing interface address or an IP pool.
IP Pool Configuration
Use Outgoing Interface Address
Use Dynamic IP Pool
Preserve Source Port
Protocol Options
default
Security Profiles
AntiVirus
not simulated yet (later)
Web Filter
not simulated yet (later)
DNS Filter
not simulated yet (later)
Application Control
not simulated yet (later)
IPS
not simulated yet (later)
File Filter
not simulated yet (later)
SSL Inspection
no-inspection
Logging Options
Log Allowed Traffic
Security Events
All Sessions
Generate Logs when Session Starts
Comments
0/1023
Enable this policy
OK
Cancel

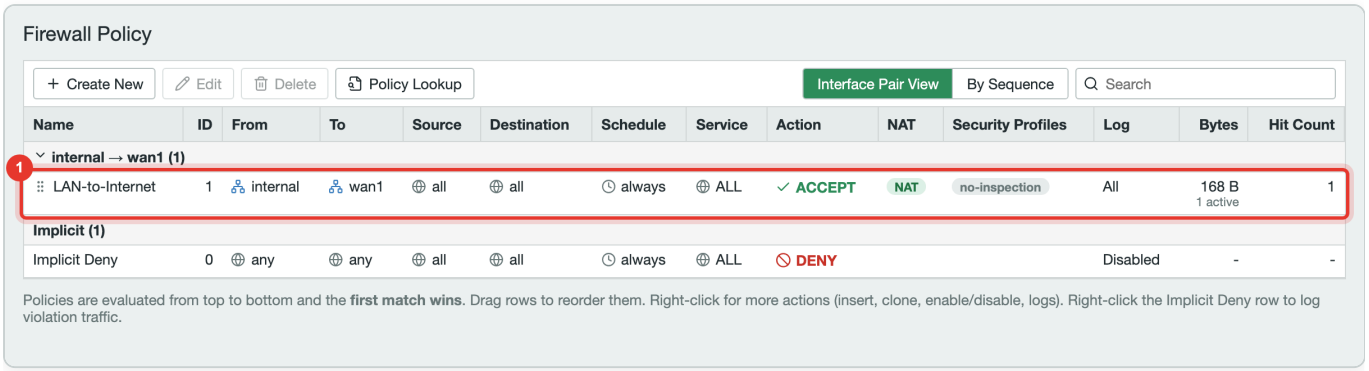
New Policy



Picker Drawer



Policy Empty



Policy List

Next: Class 3: Logs: seeing what the firewall did